

What we're doing

Can we curate at least **3,000,000** usable predicted protein-complex contacts-v1 training documents from the NVIDIA/AlphaFold Database complex release, with relaxed but calibrated confidence tiers, without allowing a handful of repeated interfaces to dominate the training mixture?

Why

The NVIDIA/AFDB complex release already contains enough usable predictions to reach three million training documents without running AlphaFold-Multimer ourselves. The earlier [#145](<https://github.com/Open-Athena/MarinFold/issues/145>) attempt became small because it was designed as a validation set: it restricted candidates to homologues of validation monomers and then selected one representative per coarse pair of independently clustered chains. For training, retaining multiple documents per **interface-aware** cluster while controlling their sampling weight should preserve the desired scale and prevent extreme redundancy.

We expect all usable high-confidence predictions to provide the core corpus. A calibrated medium-confidence tier should fill the remainder to three million. Homodimers and heterodimers, confidence tiers, and source datasets must remain separately identifiable so later training experiments can tune or ablate the mixture.

Results so far — the live census landed

The full 8.45 GB metadata mirror is normalized and censused: ****29,025,020 models**** (21,430,663 homodimers, 7,594,357 heterodimers), reproducing #145's dimer count exactly. ****1,923,625**** clear both the hard validity filter and the Tier-A quality gate.

Stage A itself is implemented and tested: lossless normalization, the yield census, exact sequence annotations with eval decontamination, deterministic Tier A/B quota selection, and a per-model terminal-status ledger.

Two findings that change the plan

****The `pilot-v0` floor of 0.5 cannot reach 3M.**** It yields 2,946,613 candidates **before** sequence-pair dedup and decontamination. The target needs a floor at or below ~ 0.4 , and realistically ~ 0.3 to leave decontamination margin. Whether that is defensible is the Stage-B pilot's job to decide.

****The 500k heterodimer floor is unreachable from AFCDB.**** Only 922,359 hard-eligible heterodimers have a nonzero ipSAE at all, and 353,774 reach $\text{ipSAE} \geq 0.1$. Clearing 500k would mean accepting models with essentially no predicted interface. The realistic ceiling is $\sim 210\text{k}$ at floor 0.30. The issue already conditioned this floor on the calibrated yield supporting it; it does not.

Archive reconnaissance: Stage D must move to the cloud

The coordinate archive is **48.8 TB** (35.20 TB homodimers, 13.58 TB heterodimers) across 17,820 referenced tars. Measured single-stream EBI throughput from the workstation is ~113 KB/s, so no variant of this runs locally.

Two properties make a selective extraction viable instead of a bulk mirror: each model is one member named `AF-<modelEntityId>-model\_v1.cif.zst`, so member names are a pure function of the model id; and EBI serves HTTP range requests, so tar headers can be walked with 512-byte reads and only selected members fetched. That moves roughly the selected payload (~1.5-2.5 TB) rather than 48.8 TB. Homodimer `chunk\_\*` tars also carry a redundant `.pdb.zst` per model that selective extraction skips entirely.

The sequence dependency, resolved

AFCDB ships no FASTA and ``modelEntityId`` is opaque, so subunit sequences come from AlphaFold DB's bulk ``sequences.fasta`` — measured at **118.0 GB**, not the 110 GB the listing rounds to. A smoke run in ``europe-west4-a`` measured **14.5 MB/s** on one 2-vCPU pod, and the same rate with 4 ranges as with 16, so the scan is Python-bound rather than network-bound. It is therefore sharded across 16 pods, each owning a byte window and subdividing it into ranges.

A sharded run cannot judge its own completeness, so ``--verify`` checks the union: that the windows tile the file contiguously and that every requested accession was matched exactly once.

Three bugs the scale found

****The header parser.**** AFDB writes `>AFDB:AF-<acc>-F1 ... UA=<acc>`, and the leading `AFDB:` defeated the `AF-...-F<n>` pattern, so every header would have produced the whole token as the "accession" and failed the missing-accession check for every row.

****EBI refuses concurrent connections.**** Seven of the first sixteen pods died on `Connection refused` from the opening HEAD. The reader now retries with backoff and jitter and resumes a dropped stream from the first unreturned byte.

****A truncated transfer reads as a clean EOF.**** Writing the reconnect test found something worse than the bug it was written for: when a server closes early, `read()` returns `b""`, so an interrupted transfer was indistinguishable from the end of the file. The range would stop short, report success, and ship a short sequence set — under-decontaminating the corpus, silently, weeks downstream. Ranges now carry the source length and treat running dry before the range end as fatal.

Guarding the join

The selector now rejects any model whose joined UniProt subunit lengths disagree with the modelled residue count ``n0chn``. Every AFCDB chain is at most 1,500 residues, well under AFDB's 2,700-residue fragmenting threshold, so an ``-F1`` model covers residues 1..L and the two should agree exactly. When they do not, the sequence we hashed, deduplicated and decontaminated on is not the sequence the document will be generated from, and the eval-leakage decision for that row is unsound.

Stage A result: 3,000,000 documents

The selector produced **3,000,000 documents**, every one a distinct sequence pair — 1,753,570 Tier A and 1,246,430 Tier B. Every one of the 29,025,020 source models carries exactly one terminal status: the ledger sums to 29,025,020 and its selected rows sum to 3,000,000.

The two dominant rejections are the relaxed quality floor (10.74M) and backbone clashes (10.44M), then eval homology (2.02M).

The heterodimer arm is exhausted, not rationed

The run selected ****183,809**** heterodimers against the reduced 200,000 floor: every Tier-A heterodimer plus every Tier-B heterodimer above the floor. The pool ran out. This is the census finding one stage deeper, now after decontamination and dedup — AFCDB does not contain 200k usable heterodimers, let alone 500k. Growing this arm needs Stage-E sources.

A bug the accounting caught

The first run's ledger summed to 3,000,017 against a 3,000,000 manifest. ****AFCDB's `modelEntityId` is not unique across its two source tables****: 130 ids appear as both a homodimer and a heterodimer, genuinely different complexes that share an identifier. Joining on the id alone duplicated ledger rows and let one source's row mask the other's in the Tier-B anti-join. Every join now keys on `(complex\_type, model\_id)`, asserted unique rather than assumed.

Why throughput needs its own sample

The stratified 50k draw spreads across 12,818 tars at **3.9 models per tar**; the real run sees **180.3**. Extraction cost is per tar, not per model — walking a tar's headers costs the same for 4 members or 400 — so timing the stratified draw would overestimate per-document cost by ~46x. A separate throughput probe takes every selected model from 20 whole tars: 4,273 models at **213.7 per tar**, the density the run actually sees.

Stage D: what the archive actually cost

16,640 of 16,640 tars walked. **2,982,771 documents of the 3,000,000 selected (99.43%)**, and the ledger closes exactly: 16,856 models whose member was not in the tar the metadata named, and 373 in 28 tars that 404. Range-addressed extraction moved ~19 GB of documents out of a 48.8 TB archive.

The run cost more than the probe predicted, and every overrun came from the same shape of bug: a per-item defect killing the unit containing it. A shared `ZstdDecompressor` across 16 threads segfaulted three runs. One TCP handshake per 512-byte read cost 9.5x. A missing tar killed 17 shards for two days over 373 documents. Preemption without per-tar resume produced nothing in 24 hours.

Stage E: PINDER supplies what AFCDB cannot

AFCDB's heterodimer pool is genuinely exhausted at **183,435** — no threshold produces more. PINDER 2024-02 was selected on indexability, not headline size: 2,319,564 systems down to 449,835, fetched **by byte offset out of a 35 GB zip** rather than downloaded.

428,221 PINDER documents, 385,354 of them on 32,939 UniProt pairs absent from AFCDB.
With AFCDB's 183,435 that is **611,656 heterodimers** — the original 500k target, met by adding a source rather than weakening a threshold.

The corpus, published

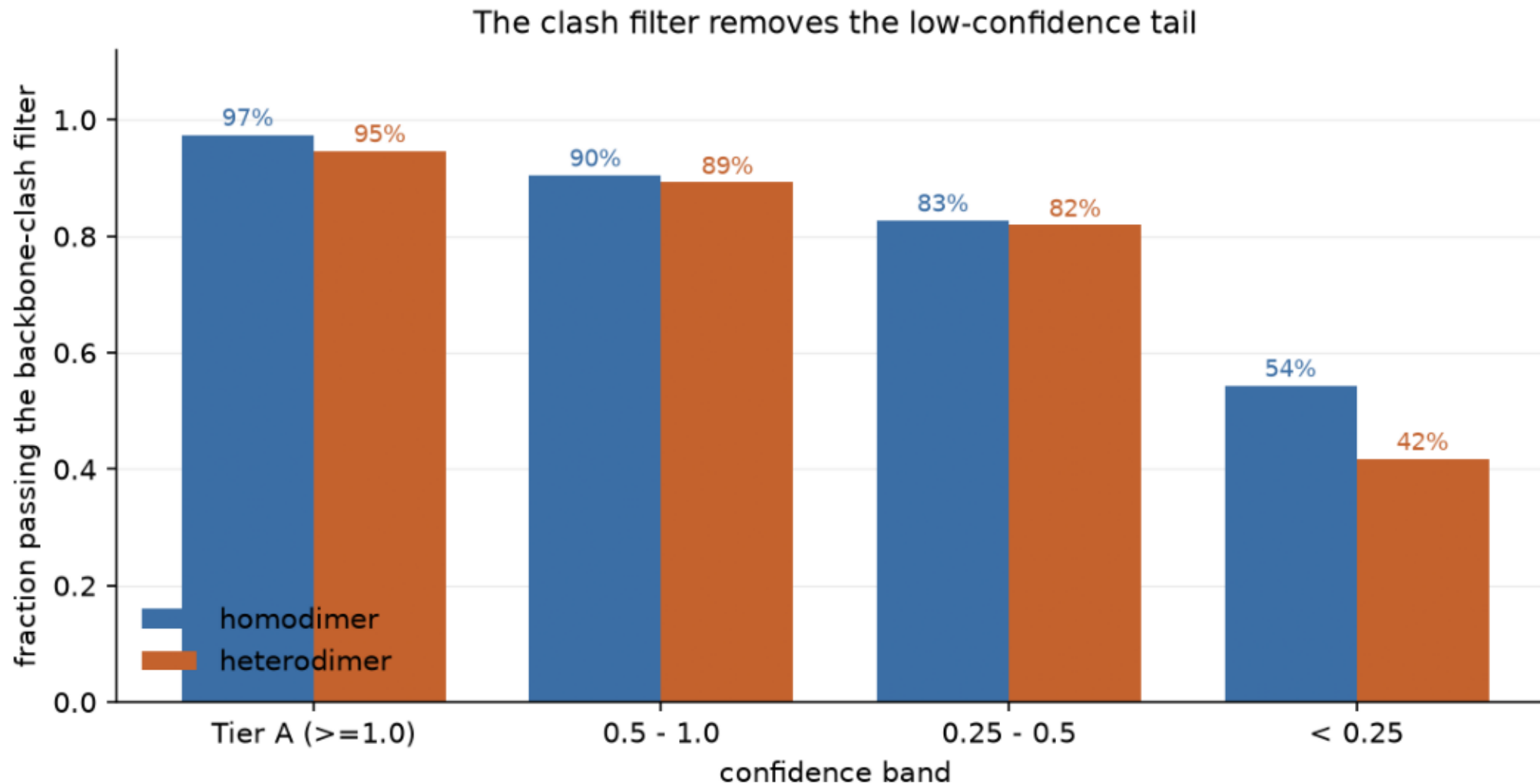
****3,410,738 documents / 11.15 B tokens**, 171 shards, 19 GB, at `open-athena/MarinFold` → `data/document\_structures/contacts\_v1\_complex/` with the tokenizer co-located.**

****261.8 M interface contacts, 12.4% of 2.11 B.**** All `sha1` distinct. Tier A 1,748,907, Tier B 1,233,610, PINDER `experimental` 428,221 — separately addressable, as the issue asks.

Redundancy is preserved in the corpus and controlled in sampling: inverse- $\sqrt{}$ cluster weighting brings the largest cluster (15,274 structures of one interface) to ****0.004%** of sampling mass**, so the 0.1% cap never binds. Interface-aware clustering exists only on the PINDER arm — AFCDB ships none and its Multimercluster data was not obtainable — so `cluster\_key` labels which kind each key is, and nothing downstream can conflate them.

clash_filter_by_band.png

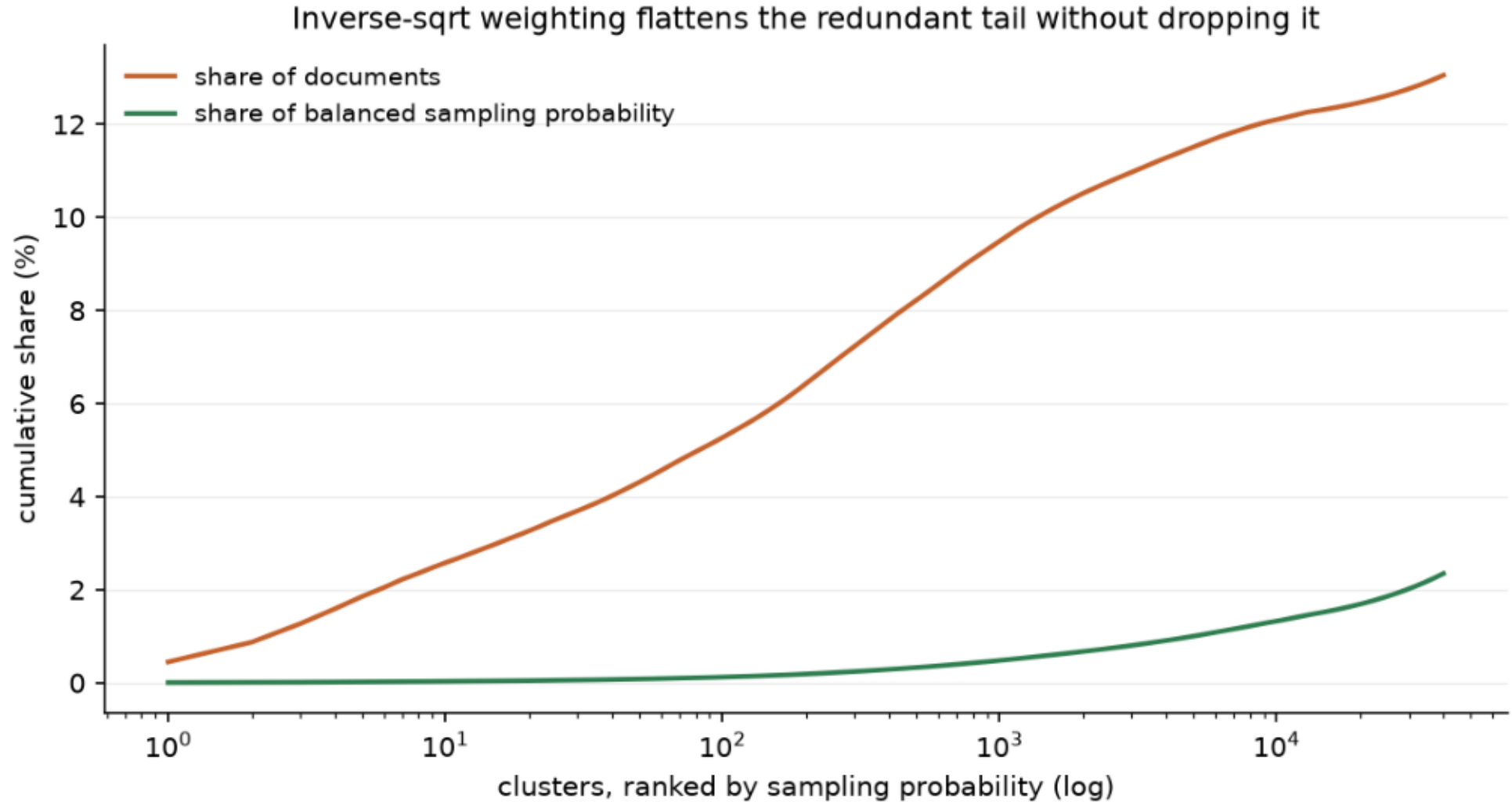
Backbone-clash pass rate by confidence band. The filter is quality-correlated rather than mis-calibrated: it keeps ~95-97% of Tier A and rejects most of the weakest band.



```
$ plots.py --input /data/exp294_predicted_complexes/metadata/normalized_*.parquet --corpus-stats-json /data/exp294_release/corpus/corpus_stats.json
```

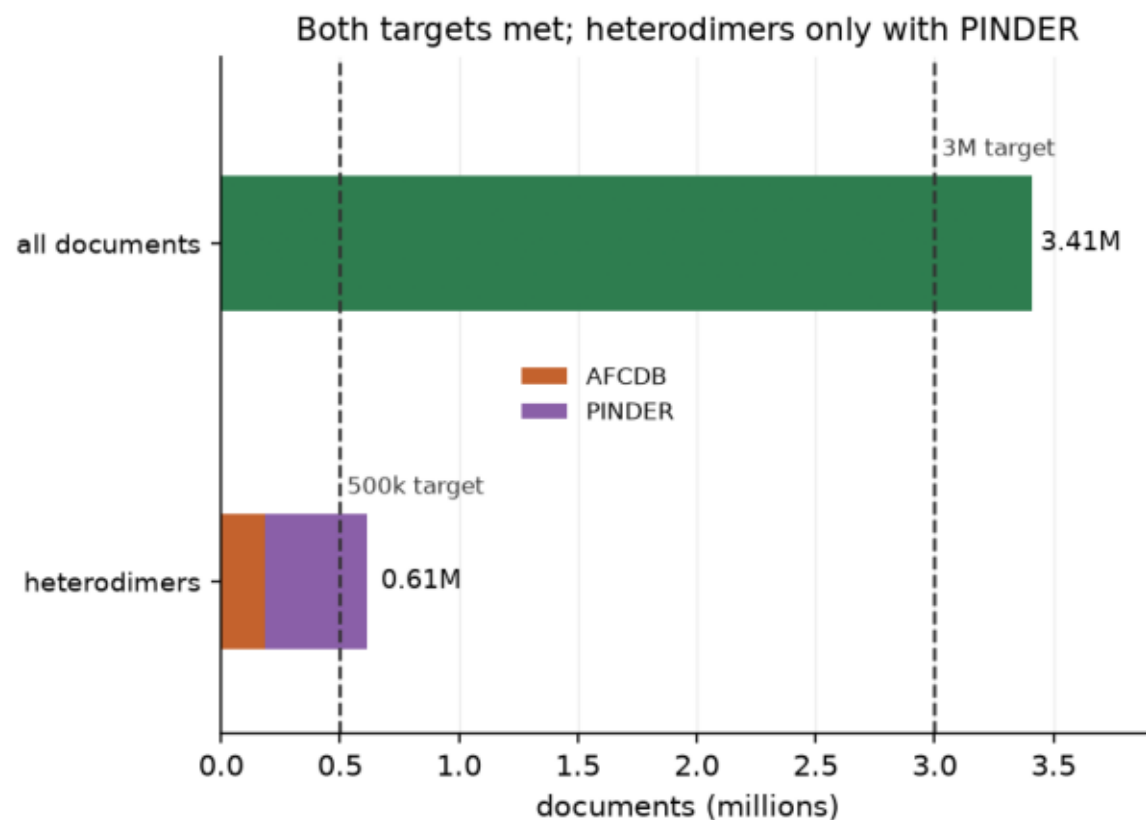
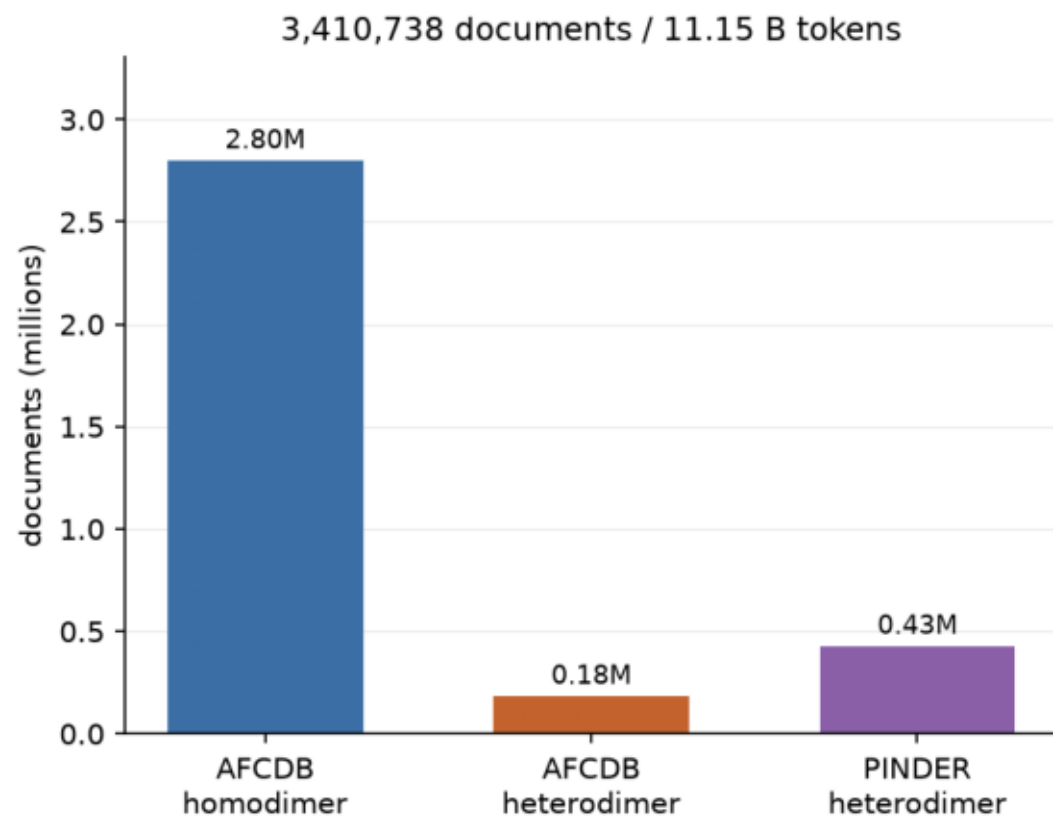
cluster_concentration.png

The top clusters are PINDER's most-crystallised interfaces. They hold a visible share of documents and far less sampling mass, so the 0.1% per-cluster cap never has to bind.



corpus_composition.png

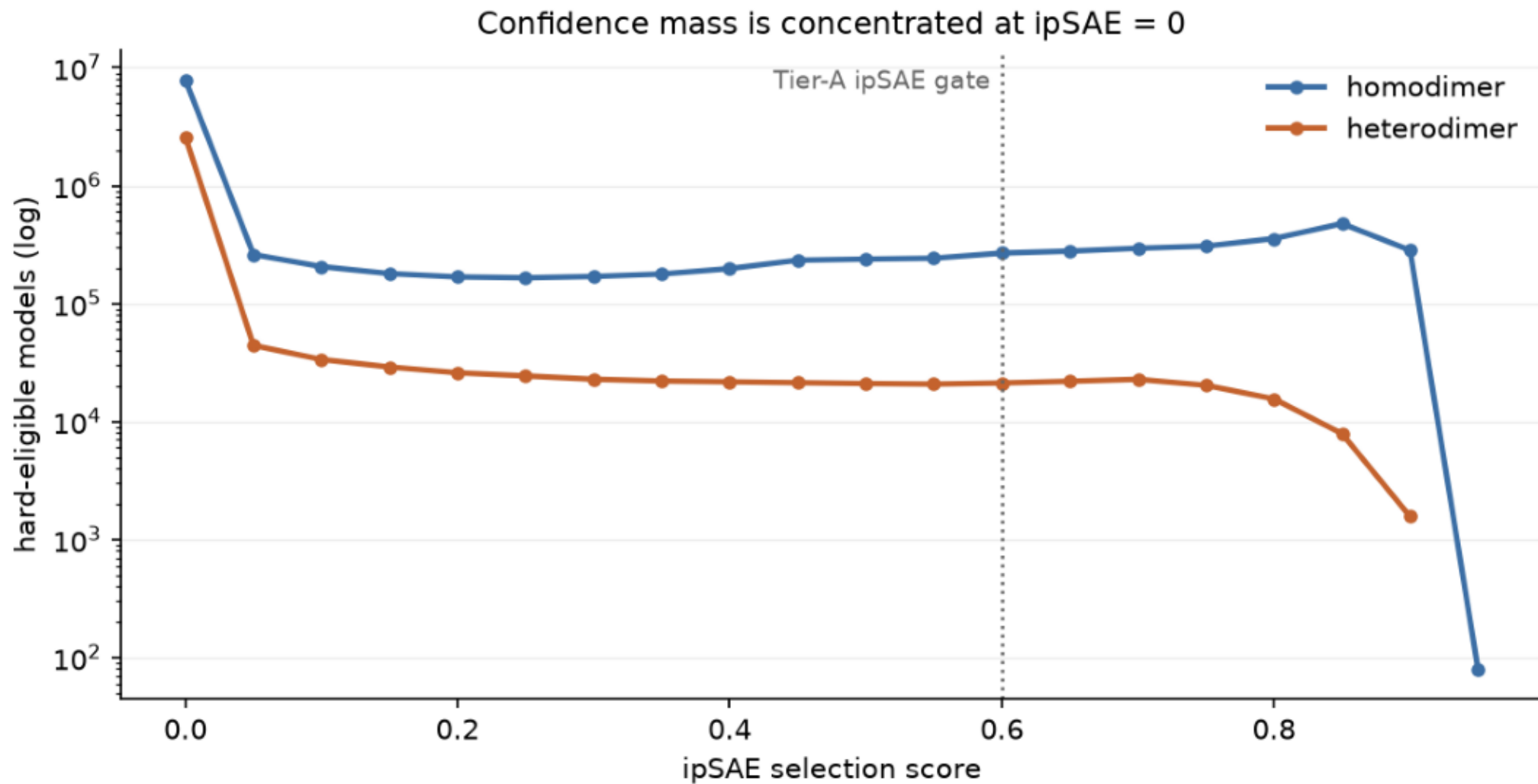
The finished corpus. AFCDB's heterodimer pool is exhausted at 183k, so the 500k heterodimer target is reached by adding PINDER.



```
$ plots.py --input /data/exp294_predicted_complexes/metadata/normalized_*.parquet --corpus-stats-json /data/exp294_release/corpus/corpus_stats.json
```

ipsae_distribution.png

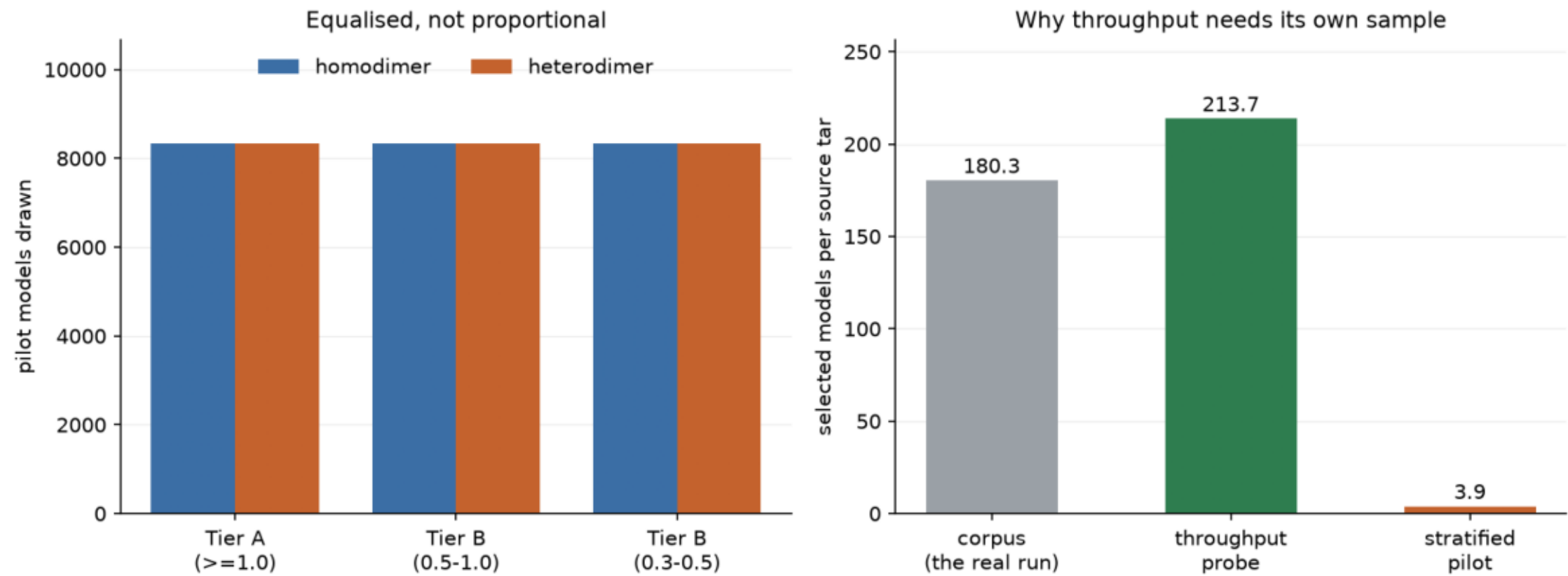
ipSAE over hard-eligible models (min over directions for homodimers, max for heterodimers; the two differ by 0.0014 on average).
76% of heterodimers score exactly 0, which is why relaxing the floor buys so little.



```
$ plots.py --input /data/exp294_predicted_complexes/metadata/normalized_*.parquet --corpus-stats-json /data/exp294_release/corpus/corpus_stats.json
```

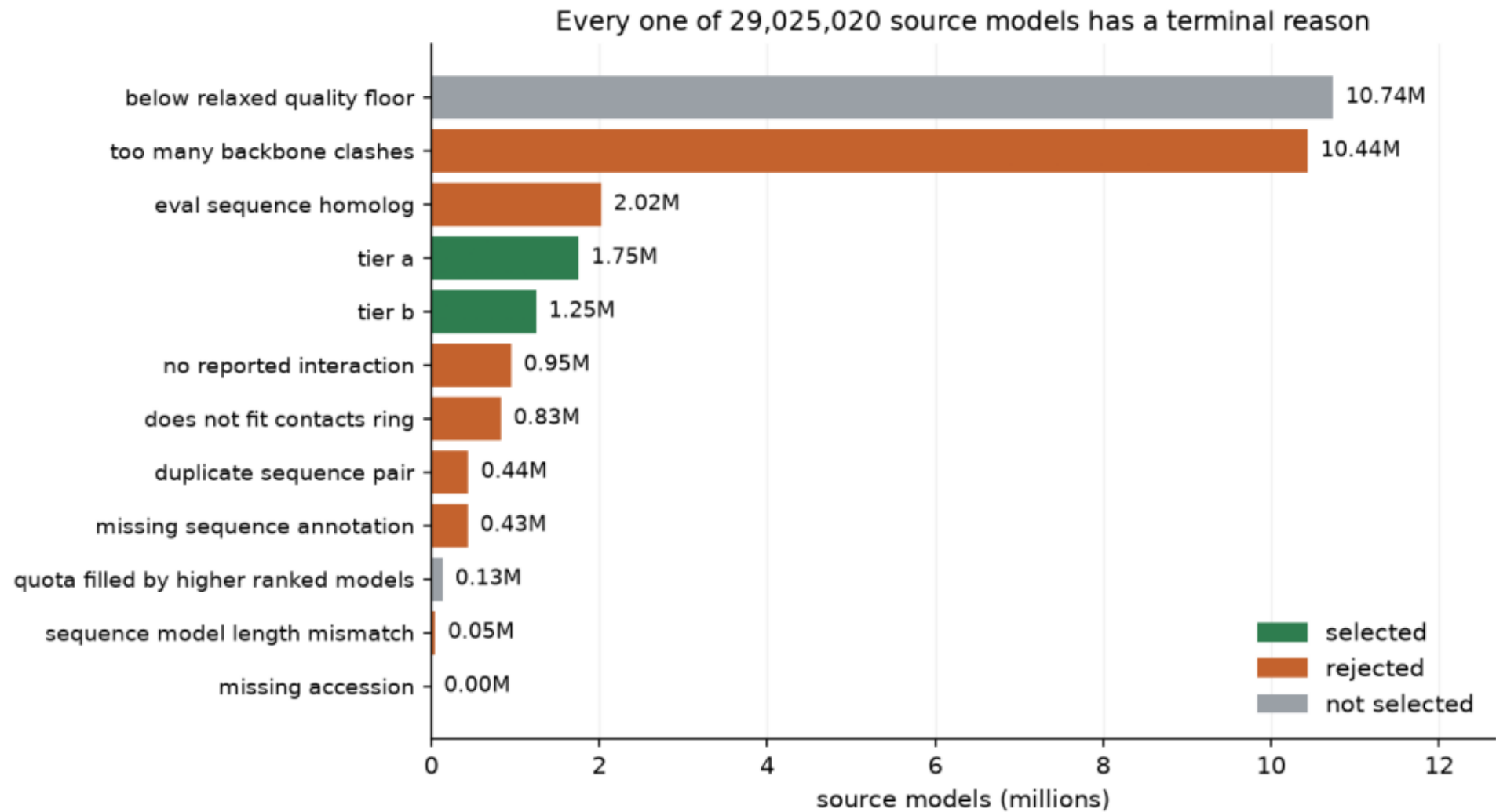
pilot_composition.png

Left: the pilot equalises across strata to over-sample the low-confidence tail. Right: extraction cost is per tar, so the stratified draw's 3.9 models/tar cannot time the run's 180.



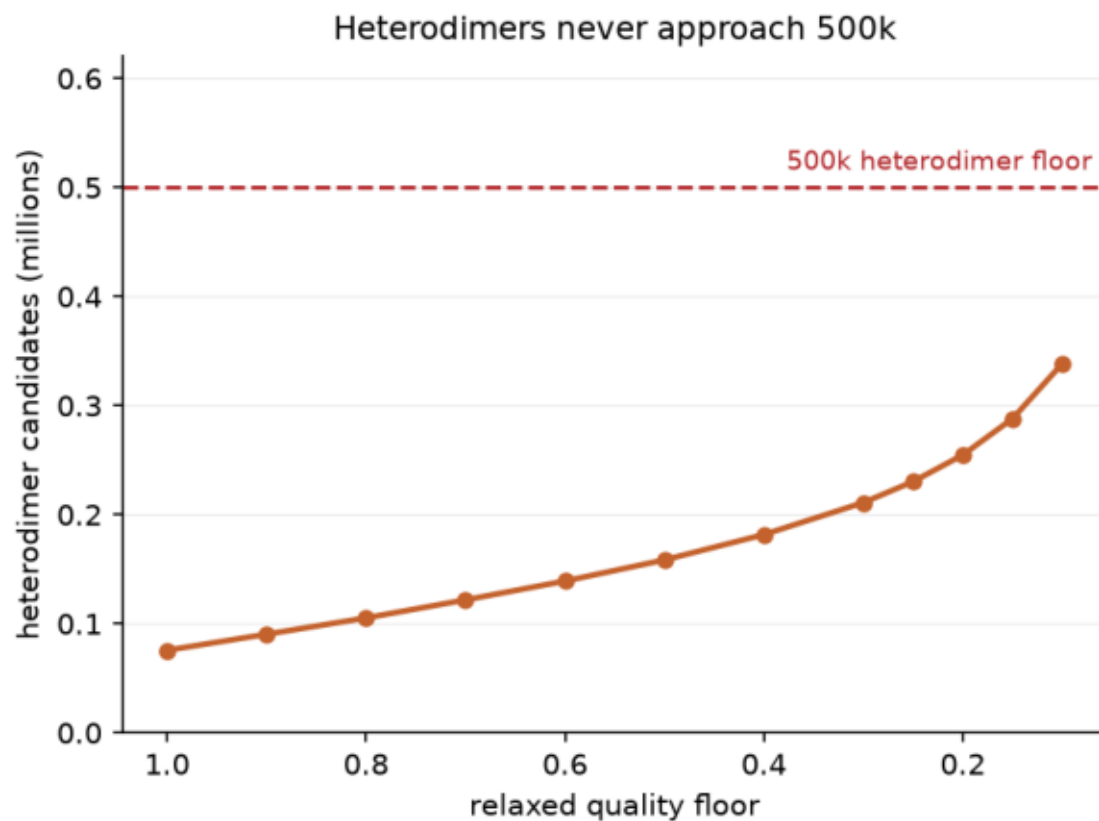
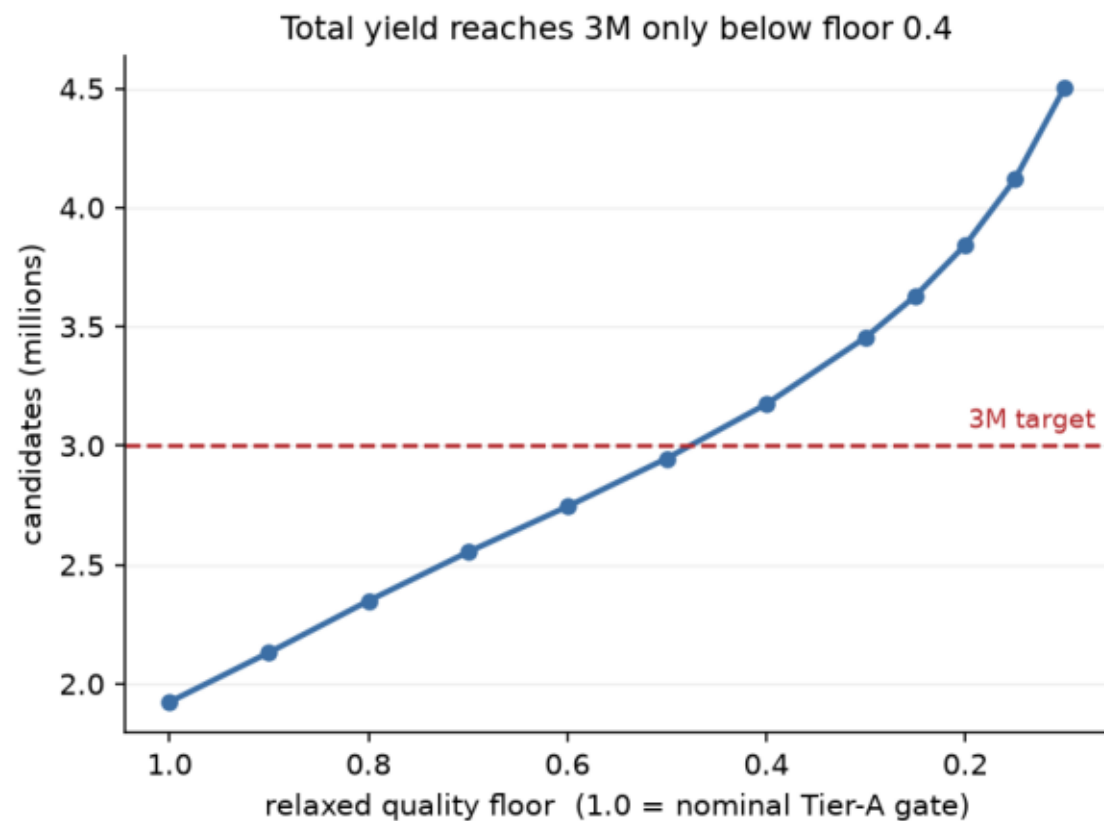
selection_funnel.png

Terminal status of every AFCDB source model. The two dominant rejections are the relaxed quality floor and backbone clashes; 3.0M are selected.



yield_frontier.png

Hard-eligible AFCDB candidates vs the relaxed quality floor. The 3M target needs a floor at or below ~0.4; 500k heterodimers is unreachable at any floor.



```
$ plots.py --input /data/exp294_predicted_complexes/metadata/normalized_*.parquet --corpus-stats-json /data/exp294_release/corpus/corpus_stats.json
```